Two-row recurrence for Koornwinder polynomials

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joint work with Lauren Williams

Conférence en l'honneur de Philippe Biane

Paris, Sorbonne Université

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Koornwinder polynomials: overview

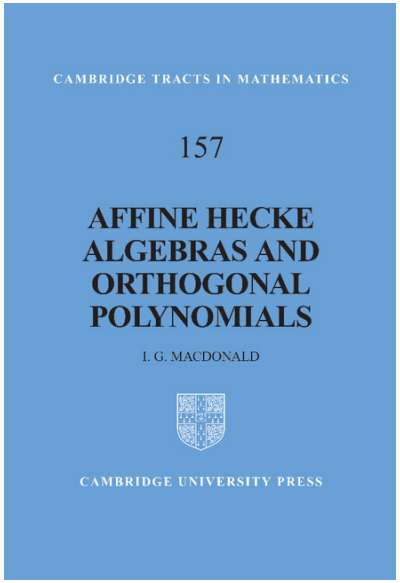
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- Koornwinder '92: symmetric polynomials, indexed by partitions, introduced as multivariable orthogonal polynomials generalizing the Askey–Wilson polynomials (one variable).
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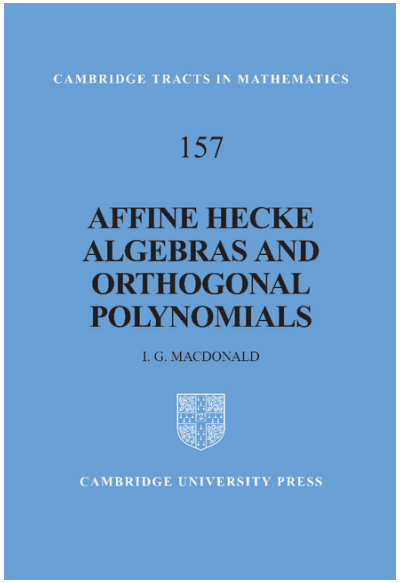


Macdonald '03

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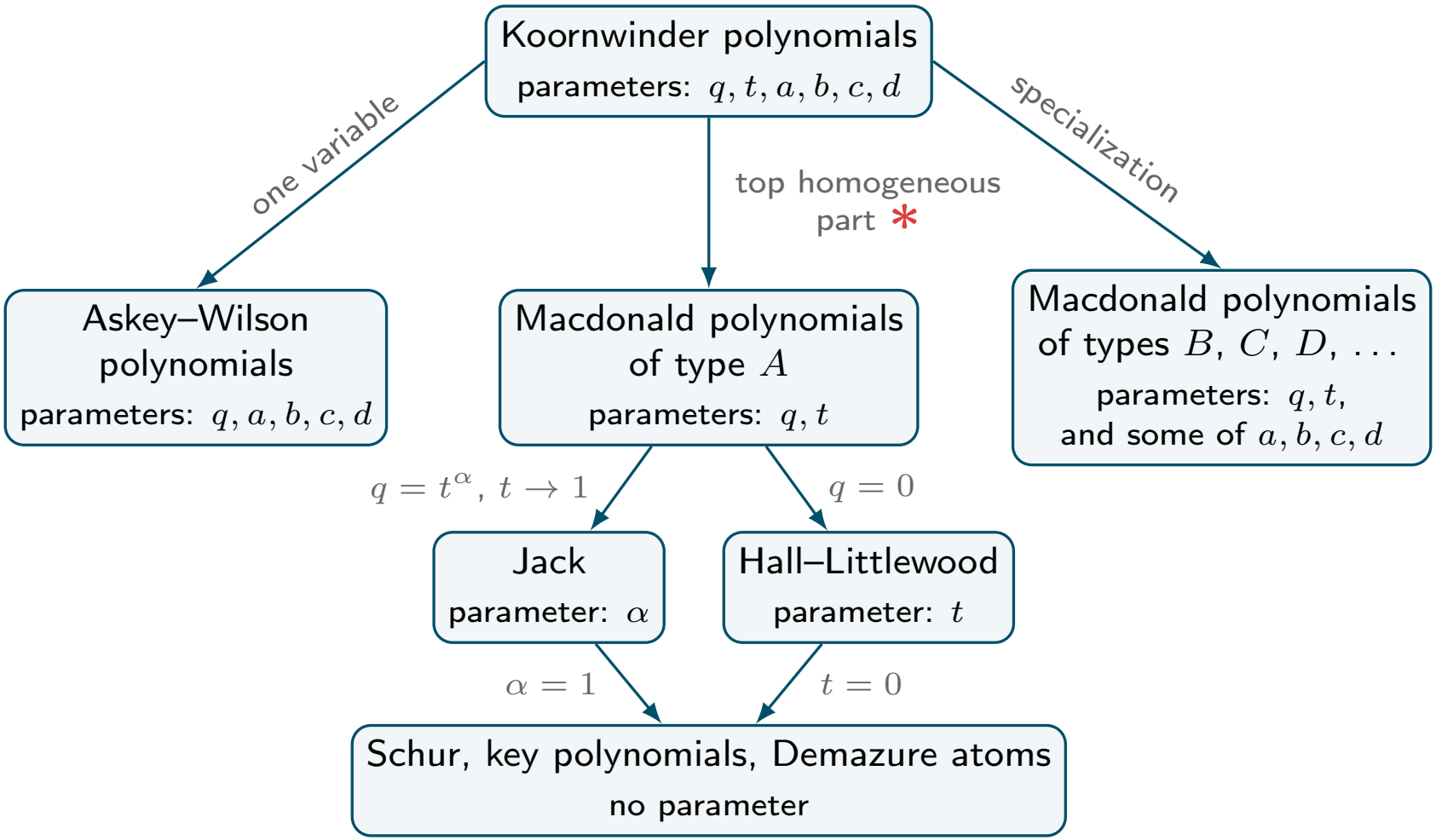
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- Cantini–Garbali–de Gier–Wheeler '16: connected Koornwinder polynomials to interacting particle systems (multispecies ASEP with open boundaries).



Macdonald '03

A family of families



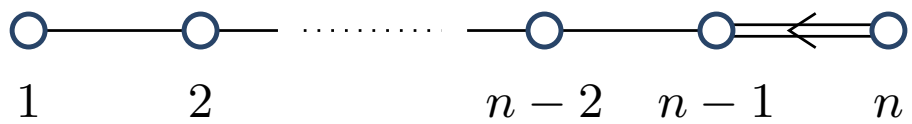
* Symmetric case: van Diejen '95. Nonsymmetric case: BD–Williams.

Notation

- A partition is $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0)$, and a composition is $\eta = (\eta_1, \ldots, \eta_n) \in \mathbb{Z}^n$.

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- $W_0 \cdot \lambda$ is the set of compositions obtained by permuting the parts of λ and changing their signs. For example,

$$W_0 \cdot (2, 0) = \{(2, 0), (-2, 0), (0, 2), (0, -2)\}.$$

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- We work with Laurent polynomials in $\mathbb{K}[z_1^{\pm 1}, \ldots, z_n^{\pm 1}]$, where $\mathbb{K} = \mathbb{Q}(q,t,a,b,c,d)$. The group W_0 acts by permuting the variables and inverting them, $z_i \mapsto z_i^{-1}$.

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Example: monomial symmetric Laurent polynomials

$$m_\lambda = \sum_{\eta \in W_0 \cdot \lambda} z^\eta, \qquad \text{e.g.} \quad m_{(2,0)} = z_1^2 + z_1^{-2} + z_2^2 + z_2^{-2}.$$

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Indexing:

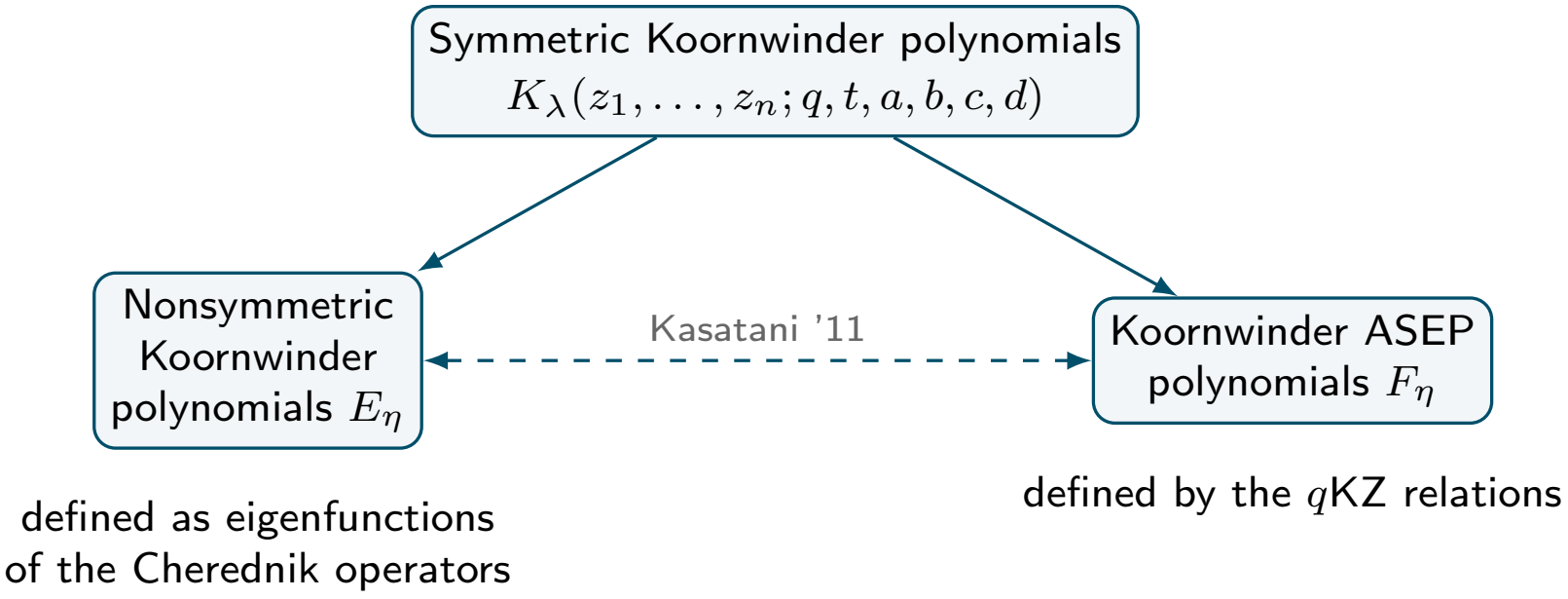
- ▷ symmetric polynomials: partitions λ,
- ▷ nonsymmetric Laurent polynomials: compositions $\eta \in \mathbb{Z}^n$,
- ▷ nonsymmetric type A polynomials (which are homogeneous): $\mu \in \mathbb{Z}^n_{\geq 0}$.

Two nonsymmetric refinements of the symmetric Koornwinder polynomials

Symmetric Koornwinder polynomials

$$K_\lambda(z_1, \ldots, z_n; q, t, a, b, c, d)$$

Two nonsymmetric refinements of the symmetric Koornwinder polynomials



Proposition (Cantini–Garbali–de Gier–Wheeler '16)

$$K_\lambda = \sum_{\eta \in W_0 \cdot \lambda} F_\eta.$$

Example

$K_{(2,0)} = F_{(2,0)} + F_{(-2,0)} + F_{(0,2)} + F_{(0,-2)}$.

Definitions of E_η and F_η

The affine Hecke algebra $\mathcal{H}$ of type $\widetilde{C}_n$ is generated by $T_0, \ldots, T_n$, with braid relations given by the diagram and $(T_i - t_i)(T_i + 1) = 0$.



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E_η: eigenfunctions of the Cherednik operators (Sahi '99, '00)

$$Y_i = T_i \cdots T_{n-1} T_n T_{n-1} \cdots T_1 \\ T_0 T_1^{-1} \cdots T_{i-1}^{-1},$$

$$Y_i E_\eta = y_i(\eta)\, E_\eta,$$

for $1 \le i \le n$, normalized by $[z^\eta] E_\eta = 1$.

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F_η: the unique family satisfying the qKZ relations (Kasatani '11), normalized by $[z^\eta] F_\eta = 1$:

$$\begin{aligned}
T_0 F_{\eta_1,\ldots} &= q^{\eta_1} F_{-\eta_1,\ldots}, && \eta_1 < 0,\\
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T_n F_{\ldots,\eta_n} &= t_n\, F_{\ldots,\eta_n}, && \eta_n = 0,\\
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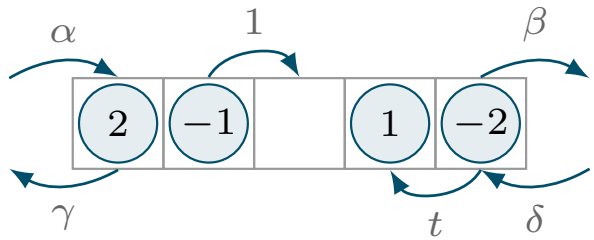
Theorem (Kasatani '11)

For the antidominant element $\eta = (-\lambda_1, \ldots, -\lambda_n)$, we have $E_\eta = F_\eta$.

Koornwinder polynomials and the multispecies ASEP

Theorem (Cantini–Garbali–de Gier–Wheeler '16)

For a, b, c, d related to the boundary rates $\alpha, \beta, \gamma, \delta$ by an explicit change of variables, the stationary probability of a state μ of the multispecies ASEP with open boundaries on n sites is proportional to $F_\mu(z_1 = \cdots = z_n = 1; q = 1, t, a, b, c, d)$.



Dynamics of the multispecies ASEP with open boundaries

The change of variables is

$$\alpha = \frac{-ac(1-t)}{(1+a)(1+c)}, \qquad \gamma = \frac{1-t}{(1+a)(1+c)},$$

$$\beta = \frac{-bd(1-t)}{(1+b)(1+d)}, \qquad \delta = \frac{1-t}{(1+b)(1+d)}.$$

Top homogeneous parts and type A

- In type A, the symmetric Macdonald polynomials have two nonsymmetric refinements, indexed by $\mu \in \mathbb{Z}^n_{\geq 0}$: the nonsymmetric Macdonald polynomials E^A_μ and the ASEP polynomials F^A_μ.
- For a Laurent polynomial f, we denote by $\mathrm{Top}(f)$ its top homogeneous part.

Theorem (BD–Williams)

For $\mu \in \mathbb{Z}^n_{\geq 0}$,

$$\mathrm{Top}(F_\mu) = F^A_\mu, \qquad \mathrm{Top}\left(E_\mu(z_1, \ldots, z_n; q, t)\right) = E^A_{\mu^{\mathrm{rev}}}(z_n, \ldots, z_1; q^{-1}, t^{-1}),$$

where $\mu^{\mathrm{rev}} = (\mu_n, \ldots, \mu_1)$.

Some examples of Koornwinder ASEP polynomials for $n = 3$

$$F_{(0,1,0)} = z_2 + \frac{t\big(t^2 ac(b+d) - a - c\big)}{1 - t^4 abcd}$$

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$$F_{(0,1,0)} = z_2 + \frac{t\big(t^2 ac(b+d) - a - c\big)}{1 - t^4 abcd}$$

$$\begin{aligned} F_{(0,1,-1)} = {} & z_2 z_3^{-1} + \frac{tbd(a+c) - b - d}{1 - t^2 abcd}\, z_2 + \frac{t\big(tac(b+d) - a - c\big)}{1 - t^2 abcd}\, z_3^{-1} \\ & + \frac{t}{(1 - t^2 abcd)(1 - t^3 abcd)} \Big((a+c)(b+d) - t\big(ac(b+d)^2 + bd(a+c)^2 - abcd\big) \\ & \quad + t^2 abcd\big((a+c)(b+d) - 1\big) - t^3 (1-t)(abcd)^2 \Big) \end{aligned}$$

In the cases of nonnegative compositions, the top degree part is represented in red. We will see that it corresponds to type A ASEP polynomials.

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$$
\begin{aligned}
F_{(0,2,0)} = {} & z_2^2 + \frac{1-t}{1-qt}\left(z_1 + qz_3\right)z_2 + \frac{q(1-t)}{1-qt}\left(z_1^{-1} + z_3^{-1}\right)z_2 + \frac{qt\left(qt^2ac(b+d) - a - c\right)}{1-q^2t^4abcd}z_2^{-1} \\
& + \frac{qt(1-t)\left(qt^2ac(b+d) - a - c\right)}{(1-qt)(1-q^2t^4abcd)}\left(z_1 + q\,z_1^{-1} + z_3 + z_3^{-1}\right) \\
& - \frac{1}{(1-qt)(1-q^2t^4abcd)}\Big(q^3t^5\left(ac(b+d) + bd(a+c)\right) + q^2t^4ac(b+d) - q^2t^3ac(b+d) \\
& \quad - q^2t^2bd(a+c) - q^2t^3(a+b+c+d) - qt^2ac(b+d) - qt^2(a+c) \\
& \quad + qt(a+c) + q(b+d) + a + c\Big)z_2 \\
& - \frac{1}{(1-qt)(1-qt^4abcd)(1-q^2t^4abcd)}\Big(q^4t^6abcd\big(t^3abcd + t^2(a+c)(b+d) \\
& \quad - t^2abcd + t^2ac + tabcd - 1\big) \\
& \quad + q^3t^4\big(t^4a^2c^2(b+d)^2 - t^4a^2bc^2d - t^3a^2b^2c^2d^2 - t^2bd(a+c)^2 - t^2ac(a+c)(b+d) \\
& \quad - t^2ac(b+d)^2 - tabcd(a+c)(b+d) + t^2abcd - ta^2bc^2d - tabcd + abcd\big) \\
& \quad - q^2t^2\big(t^4ac(a+c)(b+d) + t^3a^2c^2(b+d)^2 - t^3a^2bc^2d + t^3abcd \\
& \quad - t^2(a+c)^2 - t^2(a+c)(b+d) - t^2abcd - tbd(a+c)^2 - tac(a+c)(b+d) \\
& \quad - tac(b+d)^2 + t^2ac + tabcd - 1\big) \\
& \quad + q\big(t^3ac(a+c)(b+d) + t^4ac + t^3abcd - t(a+c)^2 - t(a+c)(b+d) \\
& \quad - t^2 + tac + t - 1\big) - tac\Big)
\end{aligned}
$$

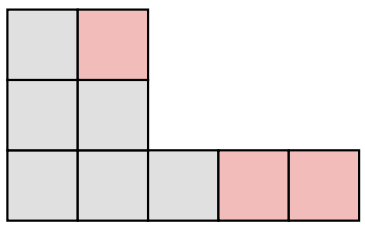
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$$F_{(0,2,1)} =$$

$$\frac{z_2^2 z_3 + \frac{(1-t)}{(1-qt^2)} z_1 z_2 z_3 + \frac{1}{(1-qt^2)(1-t^2abcd)(1-q^2t^4abcd)} \left(q^3t^8a^2b^2c^2d + q^3t^8a^2b^2c^2d^2 + q^3t^7a^2b^2cd^2 + q^3t^7ab^2c^2d^2 + q^2t^7a^2b^2c^2d + q^2t^7a^2bc^2d^2 - q^3t^7a^2bcd - q^3t^7abc^2d - q^2t^6a^2b^2c^2d - q^2t^6a^2bc^2d^2 + q^3t^6a^2bcd + q^3t^6abc^2d - q^2t^6a^2bcd - q^2t^6abc^2d - q^2t^4a^2b^2cd^2 - q^2t^4ab^2c^2d^2 + q^2t^5a^2bcd - q^2t^5ab^2cd + q^2t^5abc^2d - qt^4a^2b^2c^2d - q^2t^5abcd^2 - qt^4a^2bc^2d^2 - q^3t^5abc - q^3t^5abd - q^3t^5acd - q^3t^5bcd - q^2t^4a^2bcd - qt^5a^2bcd - q^2t^4abc^2d - qt^5abc^2d + qt^4a^2bcd + qt^4abc^2d - q^2t^4abc - q^2t^4acd + q^2t^3abc + q^2t^3acd + qt^2ab^2cd + qt^2abcd^2 - qt^3abc + q^2t^2abd - qt^3acd + q^2t^2bcd + t^2a^2bcd + t^2abc^2d + q^2t^3a + q^2t^3b + q^2t^3c + 2qt^2abc + q^2t^3d + 2qt^2acd + qt^3a + qt^3c - qta - qtc - qb - qd - a - c\right) z_2 z_3 + \frac{t(tabc+tacd-a-c)}{(1-t^2abcd)} z_2^2 + \frac{(1-t)qt^2(qt^2abc+qt^2acd-a-c)}{(1-qt^2)(1-q^2t^4abcd)} z_1 z_3 - \frac{t(1-t)}{(1-qt^2)(1-t^2abcd)(1-q^2t^4abcd)} \left(q^2t^5a^2b^2c^2d + q^2t^5a^2bc^2d^2 - q^2t^5a^2bcd - q^2t^5abc^2d + q^2t^4abc + q^2t^4acd - q^2t^3abc - q^2t^3acd + qt^3abc + qt^3acd - qt^2abc - qt^2acd - qt^2a - qt^2c - tabc - tacd + qta + qtc + a + c\right) z_1 z_2 + \frac{(1-t)q}{(1-qt^2)} z_1^{-1} z_2 z_3 + \frac{1}{(1-qt^2)(1-t^2abcd)(1-qt^3abcd)(1-q^2t^4abcd)} \left(q^4t^{10}a^3b^3c^3d^3 + q^4t^{10}a^3b^3c^2d^2 + q^4t^{10}a^3b^2c^3d^2 + q^4t^{10}a^2b^3c^3d^2 + q^4t^{10}a^3b^2c^2d^3 + q^4t^{10}a^2b^2c^3d^3 + q^3t^{10}a^3b^3c^3d + q^3t^{10}a^3b^2c^3d^2 + q^3t^{10}a^3b^3c^2d^2 + q^3t^9a^3b^2c^3d^2 - q^4t^9a^2b^2c^2d^2 - 2q^3t^8a^3b^2c^3d^2 - q^3t^8a^3b^2c^2d^3 - q^3t^9a^2b^2c^3d^2 - q^3t^9a^3bc^2d^2 - q^4t^8a^2b^2c^2d^2 - q^3t^7a^3b^3c^2d^2 - q^3t^9a^2bc^3d^2 - q^3t^7a^2b^3c^3d^2 - q^3t^7a^3b^2c^2d^3 - q^3t^7a^2b^2c^3d^3 - q^3t^8a^2b^3c^2d - q^2t^7a^3b^3c^3d - q^3t^8a^3b^2c^2d - 3q^3t^8a^2b^2c^2d^2 - q^3t^8ab^2c^3d^2 - q^2t^7a^3b^2c^3d^2 - q^3t^8a^2bc^2d^3 - q^2t^7a^3bc^3d^3 - q^4t^8a^2b^2c^2d - q^4t^8ab^2c^2d - q^2t^8a^3b^2c^2d - q^2t^8a^2b^2c^3d - q^4t^8a^2bcd^2 - q^4t^8abc^2d^2 - q^2t^8a^3bc^2d^2 - q^3t^7a^2b^2c^2d^2 - q^2t^8a^2bc^3d^2 + q^3t^8a^2bc^2d + 2q^3t^6a^2b^2c^2d^2 - q^3t^7a^2b^2c^2 - 3q^3t^7a^2bc^2d + q^2t^6a^3b^2c^2d + q^2t^6a^2b^2c^2d^2 + q^2t^6a^2b^2c^2d - q^3t^7a^2c^2d^2 + q^2t^6a^3bc^2d^2 - q^2t^6a^2b^2c^2d^2 + q^4t^7abcd + q^2t^7a^3bcd + 2q^3t^6a^2bc^2d + 2q^2t^7a^2bc^2d + q^2t^5a^2b^3c^2d + q^2t^7abc^3d + q^2t^5a^3b^2cd^2 + 4q^2t^5a^2b^2c^2d^2 + q^2t^5ab^2c^3d^2 + q^2t^5a^2bc^2d^3 - q^2t^6a^2b^2c^2 + q^3t^5a^2b^2cd + q^2t^6a^2b^2cd - 3q^2t^6a^2bc^2d + q^3t^5ab^2c^2d + q^2t^6ab^2c^2d + qt^5a^3b^2c^2d + qt^5a^2b^2c^3d + q^3t^5a^2bcd^2 + q^2t^6a^2c^2d^2 + q^3t^5abc^2d^2 + q^2t^6abc^2d^2 + qt^5a^3bc^2d^2 + qt^5a^2bc^3d^2 + q^3t^6a^2bc + q^3t^6ab^2c + q^3t^6abc^2 + 2q^2t^5a^2b^2c^2 + q^3t^6a^2bd + q^3t^6a^2cd + 3q^3t^6abcd + q^3t^6ac^2d + q^3t^6bc^2d + 3q^2t^5a^2bc^2d + qt^6a^2bc^2d + q^3t^6acd^2 + 2q^2t^5a^2c^2d^2 + qt^4a^2b^2c^2d^2 + q^3t^5abcd - qt^5a^2bc^2d + 2q^2t^5a^2bc + 2q^2t^5a^2cd - q^3t^4abcd + q^2t^5abcd - qt^4a^3bcd + 2q^2t^5ac^2d - qt^4abc^3d - 2q^2t^4a^2bc - 2q^2t^4abc^2 - 2q^2t^4a^2cd + q^2t^4abcd - qt^3a^2b^2cd - 2q^2t^4ac^2d - qt^3ab^2c^2d - qt^3a^2bcd^2 - qt^3abc^2d^2 + qt^4a^2bc - q^2t^3abc^2c + qt^4abc^2 - q^2t^3a^2bd + qt^4a^2cd - 4q^2t^3abcd - qt^4abcd + qt^4ac^2d - q^2t^3bc^2d - t^3a^2bc^2d - q^2t^3acd^2 - q^2t^4a^2 - q^2t^4ab - q^2t^4ac - q^2t^4bc - 2qt^3a^2bc - q^2t^4c^2 - 2qt^3abc^2 - q^2t^4ad - q^2t^4cd - 2qt^3a^2cd + qt^3abcd - 2qt^3ac^2d - qt^4ac - 2qt^2abcd - qt^3a^2 - qt^3ac - qt^3c^2 - q^2t^3 + 2qt^2a^2 + 2qt^2ac + 2qt^2c^2 + qtab + qtbc + qtad + qtcd + qt^2 + tac - qt + q\right) z_3 + \frac{1}{(1-qt^2)(1-t^2abcd)(1-qt^3abcd)(1-q^2t^4abcd)} \left(q^4t^{11}a^3b^3c^3d + q^4t^{11}a^3b^2c^3d^2 + q^4t^{11}a^3bc^3d^3 + q^4t^{10}a^3b^3c^2d^2 + q^4t^{10}a^2b^3c^3d^2 + q^4t^{10}a^3b^2c^3d^2 + q^4t^{10}a^3b^3c^2d^2 + q^4t^{10}a^2b^3c^2d^2 + q^4t^{10}a^3b^2c^2d^3 + q^4t^{10}a^2b^2c^3d^3 + q^3t^{10}a^3b^3c^3d + q^4t^{10}a^2b^2c^2d - q^4t^{10}a^3bc^2d^2 - q^4t^{10}a^2bc^3d^2 - q^4t^{10}ab^2c^3d^2 + q^3t^9a^3b^2c^3d^2 - q^3t^9a^3bc^3d^3 - q^4t^9a^2b^2c^2d^2 - q^3t^8a^3b^2c^2d^3 - q^3t^9a^2b^2c^3d^2 - q^3t^9a^3bc^2d^2 + q^3t^8a^2b^2c^2d^2 - q^3t^8a^3b^2c^2d^2 - q^3t^8a^2b^2c^3d^2 - q^3t^8a^2b^3c^2d^2 - q^3t^8a^3b^2c^3d^2 - q^3t^8a^2b^2c^2d^3 + q^2t^8a^3b^2c^2d^2 + q^2t^8a^2b^3c^2d^2 + q^2t^8a^2b^2c^3d^2 + q^2t^8a^2b^2c^2d^3 + q^2t^7a^3b^2c^2d^3 + q^2t^7a^2b^3c^2d^3 + q^2t^7a^2b^2c^3d^3 + q^2t^8a^2b^2c^2d^2 + q^3t^8a^2b^2c^2d^3 + q^3t^7a^2b^2c^2d^3 + q^3t^8a^2b^3c^2d^2 + q^3t^8a^3b^2c^2d^2 + q^3t^8a^2b^2c^3d^2 + q^3t^8a^2b^2c^2d^3 + q^3t^8ab^2c^3d^2 + q^2t^7a^3b^3c^3d^2 + q^3t^8a^2bc^3d^3 + q^2t^7a^3b^2c^3d^3 + q^3t^8ab^2c^2d^3 + q^3t^8a^2bc^2d^3 + q^3t^7a^2b^3c^2d^3 + q^3t^7a^3b^2c^2d^3 + q^3t^7a^2b^2c^3d^3 + q^3t^8a^2b^3c^2d^2 + q^3t^8a^3b^2c^2d^2 + q^3t^8a^2b^2c^3d^2 + q^3t^8a^2b^2c^2d^3 + q^3t^8ab^2c^2d^2 + q^3t^7a^3b^2c^2d^2 + q^3t^7a^2b^3c^2d^2 + q^3t^7a^2b^2c^3d^2 + q^3t^7a^2b^2c^2d^3 + q^3t^8a^2b^2c^2d^2 + q^2t^8a^2b^2c^2d^2 + q^2t^8a^2b^2c^2d^3 + q^2t^8a^2b^2c^3d^2 + q^2t^8a^3b^2c^2d^2 + q^2t^8a^2b^3c^2d^2 + q^2t^8a^2b^2c^3d^2 + q^2t^8a^2b^2c^2d^3 + q^2t^8ab^2c^2d^2 + q^2t^7a^3b^2c^2d^2 + q^2t^7a^2b^3c^2d^2 + q^2t^7a^2b^2c^3d^2 + q^2t^7a^2b^2c^2d^3 + q^2t^8a^2b^2c^2d^2 + qt^8a^2b^2c^2d^2 + qt^8a^2b^2c^2d^3 + qt^8a^2b^2c^3d^2 + qt^8a^3b^2c^2d^2 + qt^8a^2b^3c^2d^2 + qt^8a^2b^2c^3d^2 + qt^8a^2b^2c^2d^3 + qt^8ab^2c^2d^2 + qt^7a^3b^2c^2d^2 + qt^7a^2b^3c^2d^2 + qt^7a^2b^2c^3d^2 + qt^7a^2b^2c^2d^3 + qt^8a^2b^2c^2d^2 + t^8a^2b^2c^2d^2 + t^8a^2b^2c^2d^3 + t^8a^2b^2c^3d^2 + t^8a^3b^2c^2d^2 + t^8a^2b^3c^2d^2 + t^8a^2b^2c^3d^2 + t^8a^2b^2c^2d^3 + t^8ab^2c^2d^2 + t^7a^3b^2c^2d^2 + t^7a^2b^3c^2d^2 + t^7a^2b^2c^3d^2 + t^7a^2b^2c^2d^3 + t^8a^2b^2c^2d^2 + 2t^8a^2b^2c^2d^2 + 2t^8a^2b^2c^2d^3 + 2t^8a^2b^2c^3d^2 + 2t^8a^3b^2c^2d^2 + 2t^8a^2b^3c^2d^2 + 2t^8a^2b^2c^3d^2 + 2t^8a^2b^2c^2d^3 + 2t^8ab^2c^2d^2 + 2t^7a^3b^2c^2d^2 + 2t^7a^2b^3c^2d^2 + 2t^7a^2b^2c^3d^2 + 2t^7a^2b^2c^2d^3 +

Branching rules for Schur polynomials

$$s_\lambda(z_1, \ldots, z_n) = \sum_{\substack{\mu \subseteq \lambda \\ \lambda/\mu \text{ horizontal strip}}} z_n^{|\lambda|-|\mu|}\, s_\mu(z_1, \ldots, z_{n-1}).$$



$\lambda = (5, 2, 2)$, $\mu = (3, 2, 1)$

λ/μ is a horizontal strip: at most one box in each column.

Branching rules for Macdonald polynomials

$$P_\lambda(z_1,\dots,z_n;q,t) = \sum_{\substack{\mu\subseteq\lambda \\ \lambda/\mu \text{ horizontal strip}}} \psi_{\lambda/\mu}(q,t)\, z_n^{|\lambda|-|\mu|} P_\mu(z_1,\dots,z_{n-1};q,t),$$

with branching coefficients (Macdonald '95)

$$\psi_{\lambda/\mu}(q,t) = \prod_{s\in R_{\lambda/\mu}\setminus C_{\lambda/\mu}} \frac{b_\mu(s)}{b_\lambda(s)}, \qquad b_\lambda(s) = \frac{1-q^{a_\lambda(s)}t^{l_\lambda(s)+1}}{1-q^{a_\lambda(s)+1}t^{l_\lambda(s)}},$$

where $R_{\lambda/\mu}$ (resp. $C_{\lambda/\mu}$) is the union of rows (resp. columns) meeting λ/μ.

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Branching rules also exist for

- interpolation Macdonald polynomials (Okounkov '98),
- symmetric Koornwinder polynomials (van Diejen–Emsiz '15).

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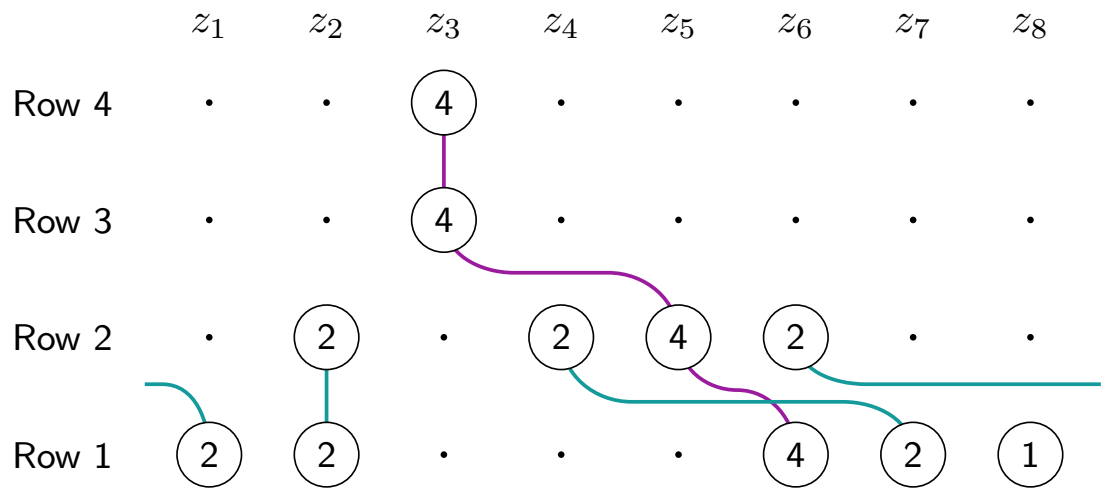
Branching rules also exist for

- interpolation Macdonald polynomials (Okounkov '98),
- symmetric Koornwinder polynomials (van Diejen–Emsiz '15).

These branching rules have two limitations:

- they do not have a direct analogue for nonsymmetric polynomials;
- except for the Schur case, the branching coefficients are complicated rational expressions in q and t, and the resulting formulas do not make integrality and positivity manifest.

Type A: multiline queues



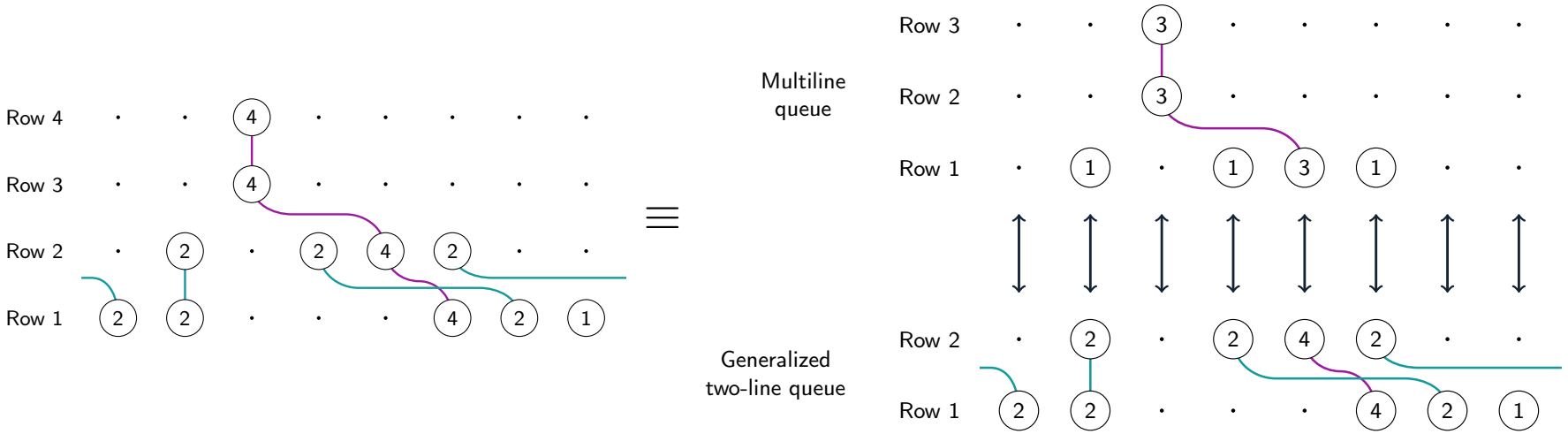
A multiline queue (MLQ) with 4 rows, of type $(2, 2, 0, 0, 0, 4, 2, 1)$.

- The type of an MLQ is given by the labels of its bottom row (0 for an empty position).
- Ball weight: each ball in column i contributes z_i; here $z_1 z_2^2 z_3^2 z_4 z_5 z_6^2 z_7 z_8$.
- Pairing weight: a rational function in q, t, multiplicative over the layers: here $\mathrm{wt}_{\mathrm{pairing}} = \mathrm{wt}_1\, \mathrm{wt}_2\, \mathrm{wt}_3$.

Theorem (Corteel–Mandelshtam–Williams '22)

For $\kappa \in \mathbb{Z}^n_{\geq 0}$, F^A_κ is the generating function of MLQs of type κ.

Type A two-row recurrence

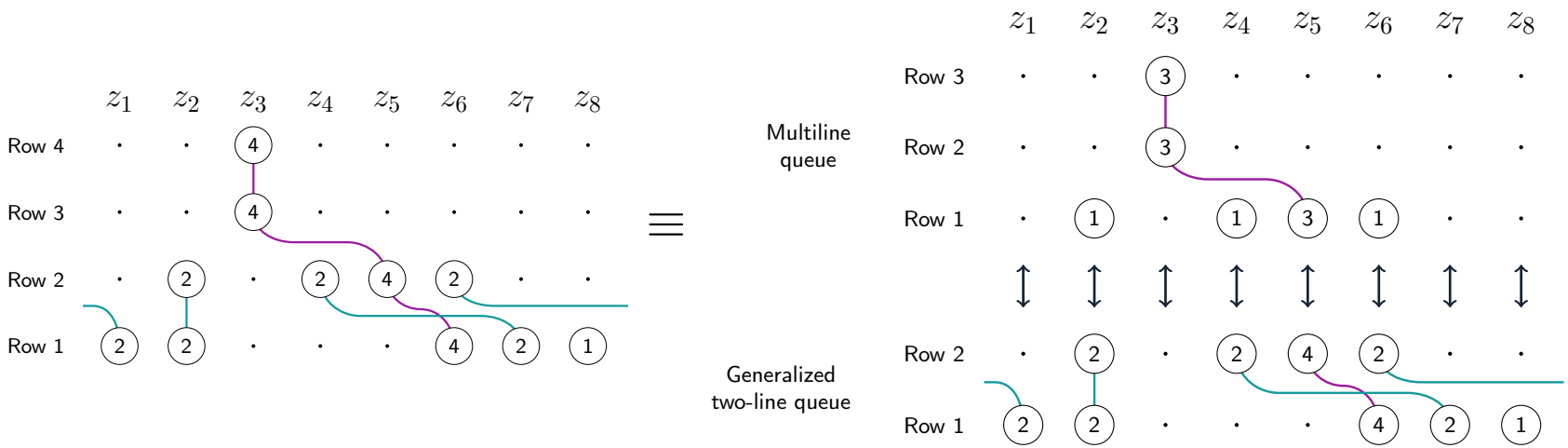


Since the pairing weight is multiplicative over layers, removing the bottom layer gives

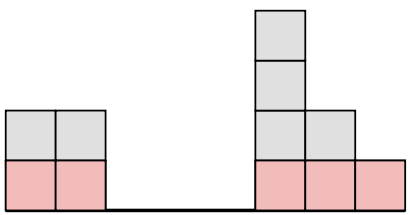
$$F_\kappa^A(z_1,\dots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\dots,z_n;q,t),$$

where $\kappa_i^- = \max\{\kappa_i - 1, 0\}$, and $c_\kappa^\mu(q,t)$ is the generating function of generalized two-line queues.

Type A two-row recurrence

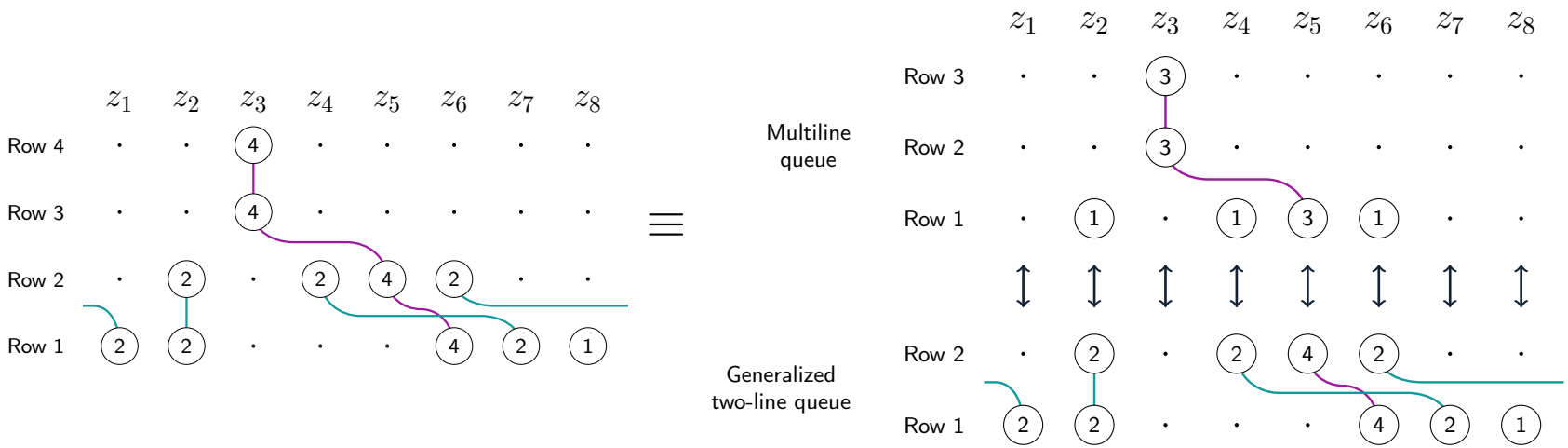


$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$

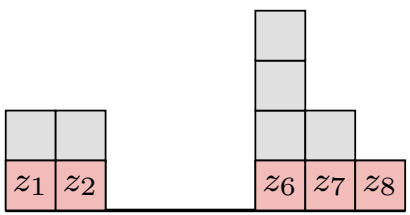


$\kappa = (2,2,0,0,0,4,2,1)$

Type A two-row recurrence

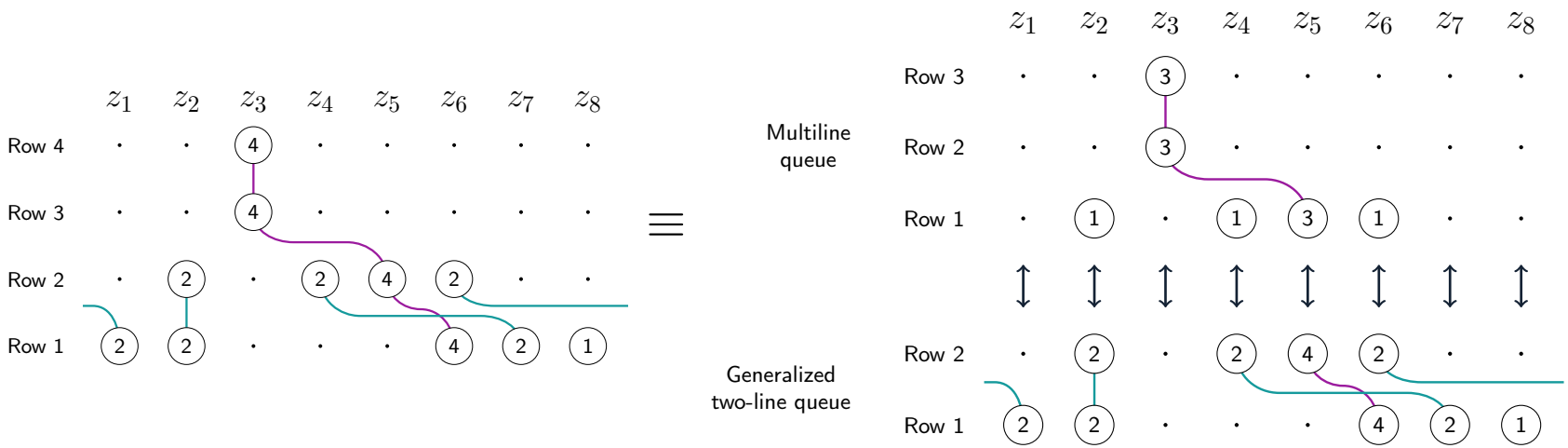


$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$

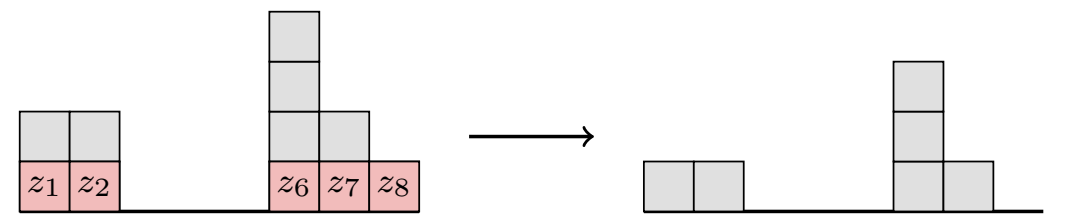


$\kappa = (2,2,0,0,0,4,2,1)$

Type A two-row recurrence

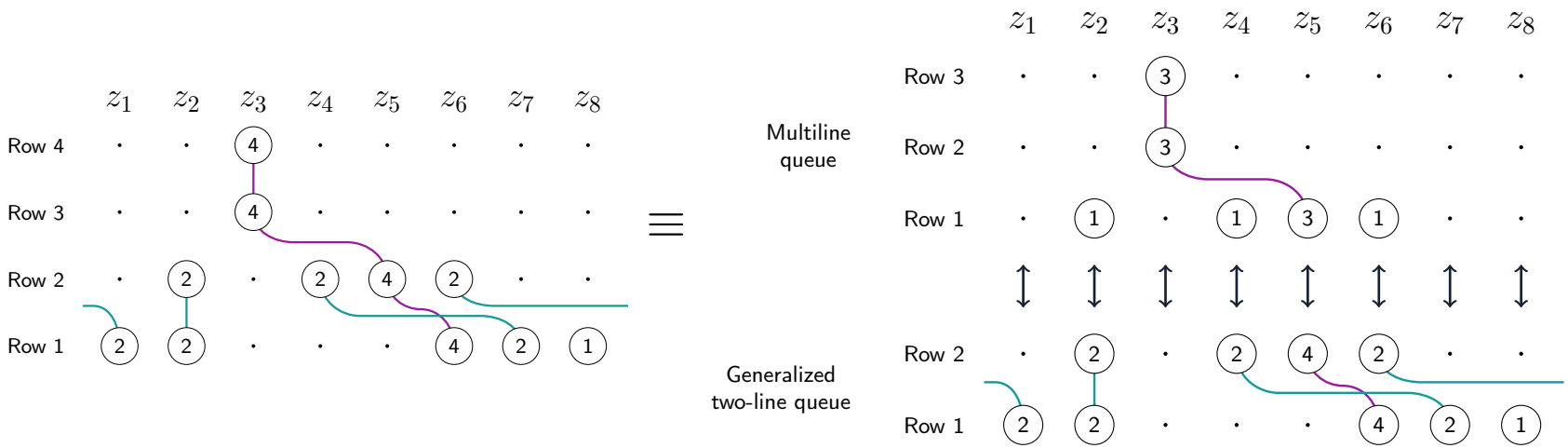


$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$

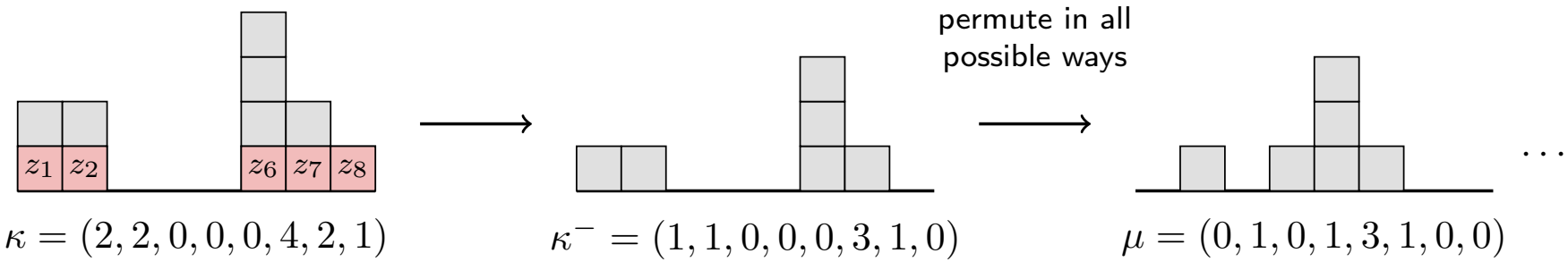


$\kappa = (2,2,0,0,0,4,2,1)$ $\qquad$ $\kappa^- = (1,1,0,0,0,3,1,0)$

Type A two-row recurrence



$$F_\kappa^A(z_1,\ldots,z_n;q,t) = \prod_{i:\,\kappa_i>0} z_i \sum_{\mu\in S_n\cdot\kappa^-} c_\kappa^\mu(q,t)\, F_\mu^A(z_1,\ldots,z_n;q,t)$$



Remark. One may think of this as a *transposed* branching rule: instead of removing one column, we remove one row.

Two-row recurrence for Koornwinder polynomials

Theorem A (two-row recurrence)

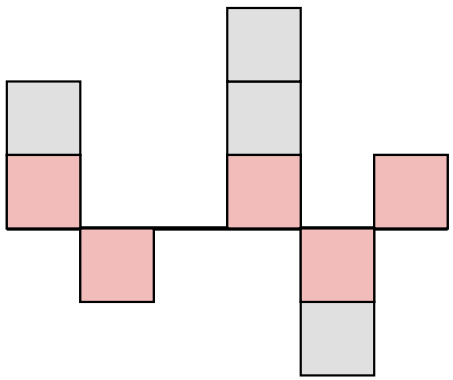
For $\kappa \in \mathbb{Z}^n$, *set* $(\kappa^-)_i = \kappa_i - \mathrm{Sign}(\kappa_i)$. *Then*

$$F_\kappa(z_1,\dots,z_n;q,t,a,b,c,d) = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, z^\nu F_\mu(z_1,\dots,z_n;q,t,a,b,c,d),$$

where $\mu \in W_0 \cdot \kappa^-$ *and* $\nu \in \mathbb{Z}^n$ *with* $\nu_i \in \{0, \mathrm{Sign}(\kappa_i)\}$.

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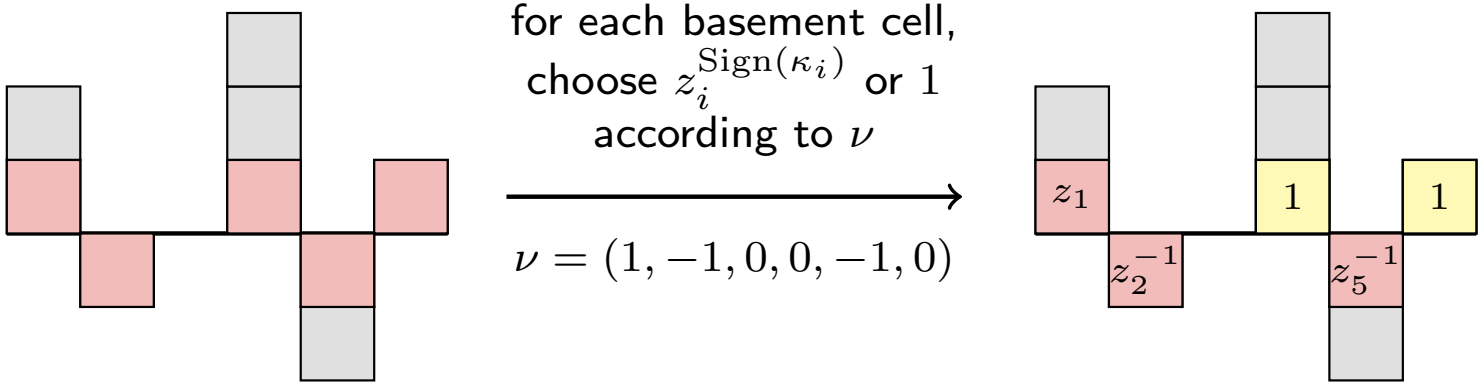
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$\kappa = (2,-1,0,3,-2,1)$

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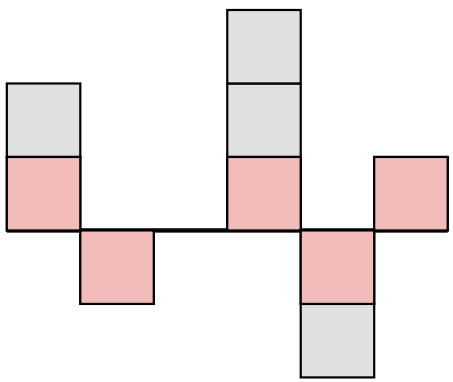


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$z^\nu = z_1 z_2^{-1} z_5^{-1}$

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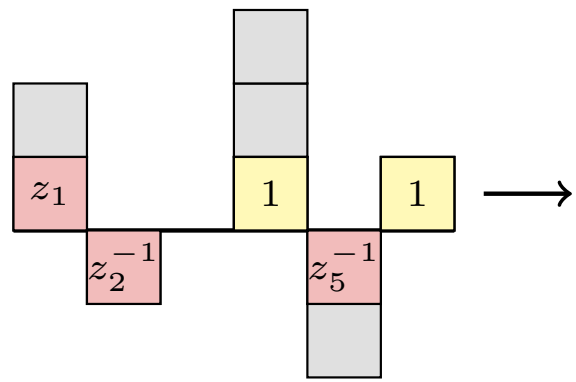
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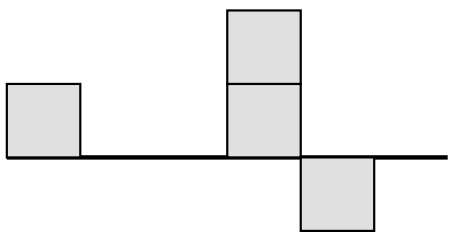
$\kappa = (2,-1,0,3,-2,1)$

for each basement cell, choose $z_i^{\mathrm{Sign}(\kappa_i)}$ or 1 according to ν

$\nu = (1,-1,0,0,-1,0)$



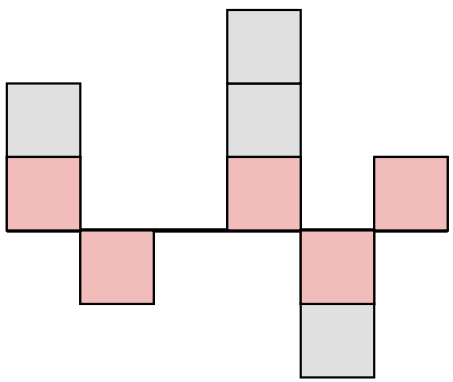
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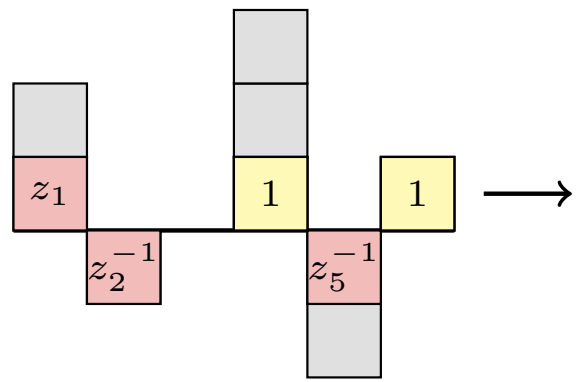
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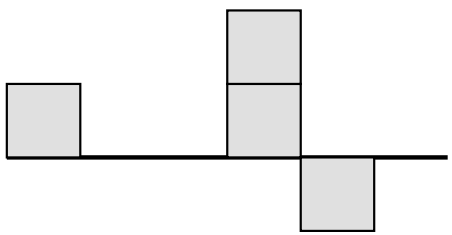
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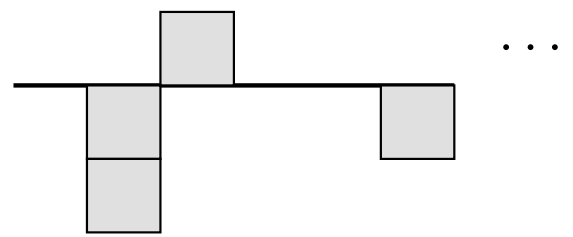


$z^\nu = z_1 z_2^{-1} z_5^{-1}$



$\kappa^- = (1,0,0,2,-1,0)$

permute and flip signs in all possible ways



$\dots$

$\mu = (0,-2,1,0,0,-1)$

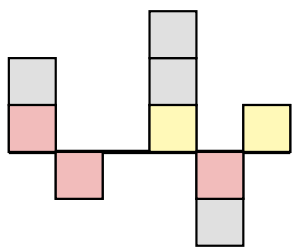
Two-line configurations

For $\kappa \in \mathbb{Z}^n$, $\nu \in \mathbb{Z}^n$ with $\nu_i \in \{0, \mathrm{Sign}(\kappa_i)\}$, and $\mu \in W_0 \cdot \kappa^-$, we write

$$a_\kappa^\mu(\nu) = \left[\begin{array}{c} \sigma \\ \tau \end{array} \right].$$

τ is obtained from κ by putting a bar on κ_i when $\nu_i = 0$.

Example: $\kappa = (2, -1, 0, 3, -2, 1)$, $\nu = (1, -1, 0, 0, -1, 0)$.



$$\tau = (2, -1, 0, \overline{3}, -2, \overline{1})$$

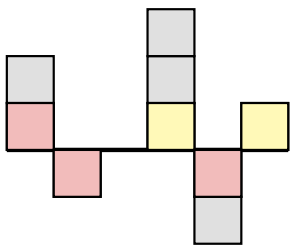
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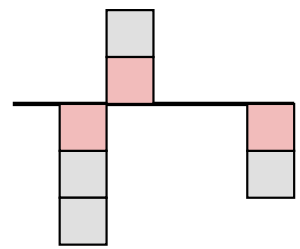
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$\tau = (2, -1, 0, \overline{3}, -2, \overline{1})$

$\sigma = \mu \oplus \mathrm{Sign}(\mu)$ is obtained from μ by adding a basement cell to each nonzero column.

Example: $\mu = (0, -2, 1, 0, 0, -1)$.



$\sigma = (0, -3, 2, 0, 0, -2)$

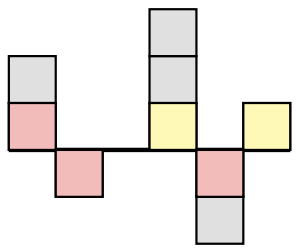
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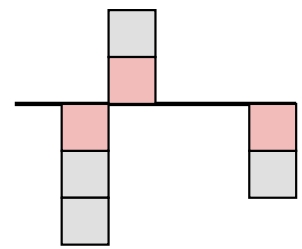
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Example: $\mu = (0, -2, 1, 0, 0, -1)$.



$$\sigma = (0, -3, 2, 0, 0, -2)$$

$$a_\kappa^\mu(\nu) = \begin{bmatrix} 0 & -3 & 2 & 0 & 0 & -2 \\ 2 & -1 & 0 & \overline{3} & -2 & \overline{1} \end{bmatrix}$$

Two-line configurations

Fix a partition λ. The recurrence coefficients are encoded by the diagrams

$$\begin{bmatrix} \sigma \\ \tau \end{bmatrix},$$

where

- τ is a signed permutation of λ, potentially with some bars,
- σ is a signed permutation of λ with its parts equal to 1 replaced by 0.

Example: for $\lambda = (3, 2, 2, 1, 1, 0)$,

$$\begin{bmatrix} 0 & -3 & 2 & 0 & 0 & -2 \\ 2 & -1 & 0 & \overline{3} & -2 & \overline{1} \end{bmatrix}.$$

Main proof idea

- Use the double affine Hecke algebra of type $(C^\vee, C)$ (Sahi '99) to define a refinement of the polynomial representation, in which we have a formal copy $\mathsf{Z}_1, \ldots, \mathsf{Z}_n$ of the variables $z_1, \ldots, z_n$.
- We then define refined polynomials $\widehat{E}_\kappa$ and $\widehat{F}_\kappa$, with

$$\widehat{F}_\kappa = \sum_{(\nu,\mu)} a_\kappa^\mu(\nu)\, \mathsf{Z}^\nu F_\mu.$$

- Replacing Z_i by z_i sends $\widehat{E}_\kappa \mapsto E_\kappa$ and $\widehat{F}_\kappa \mapsto F_\kappa$, which gives Theorem A.

Local rules

Theorem B

The two-line configurations are characterized by a family of local rules.

Only the columns involved are represented; the other columns are the same on both sides. Here $m > \ell$ and $k > 0$.

In the bulk (two consecutive columns):

$$\begin{bmatrix} m & \ell \\ -k & 0 \end{bmatrix} = t \begin{bmatrix} \ell & m \\ 0 & -k \end{bmatrix} + (1-t) \begin{bmatrix} m & \ell \\ 0 & -k \end{bmatrix}.$$

On the boundary (first column):

$$\begin{bmatrix} 0 & \cdots \\ \overline{k} & \cdots \end{bmatrix} = -q^{k-1}ac \begin{bmatrix} 0 & \cdots \\ \overline{-k} & \cdots \end{bmatrix} - q^{k-1}(a+c) \begin{bmatrix} 0 & \cdots \\ -k & \cdots \end{bmatrix}.$$

In total, there are around 90 local rules (43 up to symmetry).

A cyclic algorithm

Each local rule transports an active column to the next position, plus smaller terms, for some order on two-line configurations.

$$\begin{bmatrix} m & \ell \\ -k & 0 \end{bmatrix} = t \begin{bmatrix} \ell & m \\ 0 & -k \end{bmatrix} + \underbrace{(1-t)\begin{bmatrix} m & \ell \\ 0 & -k \end{bmatrix}}_{\text{smaller terms}},$$

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Each local rule transports an active column to the next position, plus smaller terms, for some order on two-line configurations.
We solve the system by induction, moving the active column cyclically:

$$C = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -2 \end{bmatrix}$$

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$$(1 - qt^2)\, C = \underbrace{(1-t) \begin{bmatrix} 2 & 0 & 0 \\ 1 & 0 & -2 \end{bmatrix} + t(1-t) \begin{bmatrix} 0 & 0 & -2 \\ 1 & 0 & -2 \end{bmatrix}}_{\text{smaller terms}}.$$

Application 1: nonsymmetric Askey–Wilson polynomials

Theorems A and B give explicit recurrences in the one-variable case ($n = 1$):

Theorem

For $k \geq 1$,

$$F_{(k+1)} = \big(z + u_k\big)F_{(k)} + \big(v_k z + w_k\big)F_{(-k)},$$
$$F_{(-k-1)} = \big(\widetilde{v}_k z^{-1} + \widetilde{w}_k\big)F_{(k)} + \big(z^{-1} + \widetilde{u}_k\big)F_{(-k)},$$

with $u_k = w_k\widetilde{v}_k$, $\widetilde{u}_k = v_k\widetilde{w}_k$, *and*

$$v_k = q^{k-1}\frac{ac + q - acq^k - abcdq^k}{1 - abcdq^{2k-1}}, \qquad w_k = q^k\frac{-(a+c) + ac(b+d)q^k}{1 - abcdq^{2k}},$$
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This result refines the three-term relation for symmetric Askey–Wilson polynomials $P_k = F_{(k)} + F_{(-k)}$.

All these coefficients are rational functions of q^k. In particular, for $n = 1$, all two-line configurations are given by 8 rational functions of q^k. For any n, there are only finitely many rational functions to compute.

Application 2: top homogeneous parts and type A

Definition

A two-line configuration is simple if all its labels are regular (no bars), and two nonzero labels in the same column have the same sign.

Example

$$\begin{bmatrix} 2 & -2 & 0 & 0 & 3 & 0 \\ 3 & -2 & 1 & -1 & 0 & 2 \end{bmatrix} \text{ is simple,}$$

$$\begin{bmatrix} 0 & -3 & 2 & 0 & 0 & -2 \\ 2 & -1 & 0 & \overline{3} & -2 & \overline{1} \end{bmatrix} \text{ is not.}$$

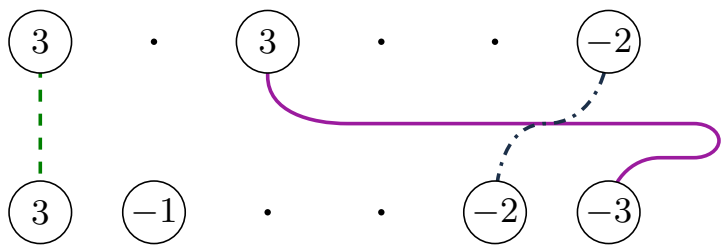
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A simple two-line configuration $\begin{bmatrix} \sigma \\ \tau \end{bmatrix}$ *is the generating function of reflected two-line queues with top row* σ *and bottom row* τ*.*

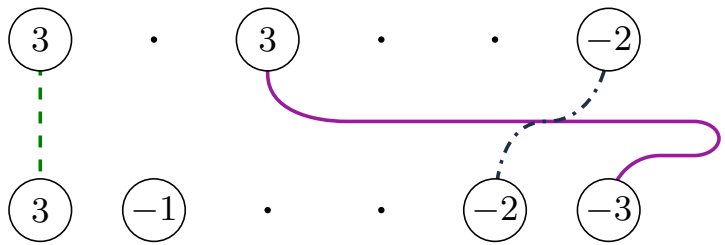


$$\begin{bmatrix} 3 & 0 & 3 & 0 & 0 & -2 \\ 3 & -1 & 0 & 0 & -2 & -3 \end{bmatrix}$$

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Together with the two-row recurrence (Theorem A), this gives:

Theorem C (top homogeneous parts)

For $\mu \in \mathbb{Z}^n_{\geq 0}$,

$$\mathrm{Top}(F_\mu) = F^A_\mu, \qquad \mathrm{Top}\left(E_\mu(z_1,\ldots,z_n;q,t)\right) = E^A_{\mu^{\mathrm{rev}}}(z_n,\ldots,z_1;q^{-1},t^{-1}).$$

Main results (BD–Williams)

Theorem A (two-row recurrence)

For $\kappa \in \mathbb{Z}^n$, *set* $(\kappa^-)_i = \kappa_i - \mathrm{Sign}(\kappa_i)$. *Then*

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where $\mu \in W_0 \cdot \kappa^-$ *and* $\nu \in \mathbb{Z}^n$ *with* $\nu_i \in \{0, \mathrm{Sign}(\kappa_i)\}$.

Theorem B (local rules)

The coefficients $a_\kappa^\mu(\nu)$ *are uniquely characterized by a family of local rules.*

Theorem C (top homogeneous parts)

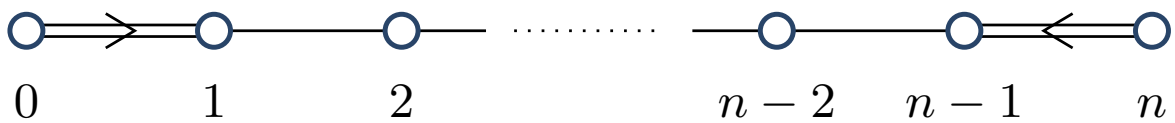
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The affine Weyl group of type $\widetilde{C}_n$

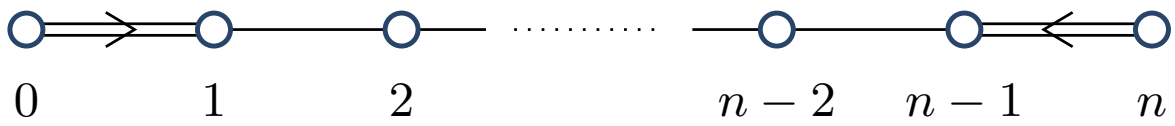
The affine Weyl group W of type $\widetilde{C}_n$ is generated by $s_0, s_1, \ldots, s_n$ with relations

$$\begin{aligned} s_i s_j &= s_j s_i, && |i-j| \geq 2, \\ s_i s_{i+1} s_i &= s_{i+1} s_i s_{i+1}, && 1 \leq i < n-1, \\ s_0 s_1 s_0 s_1 &= s_1 s_0 s_1 s_0, \\ s_{n-1} s_n s_{n-1} s_n &= s_n s_{n-1} s_n s_{n-1}, \\ s_i^2 &= 1, && 0 \leq i \leq n. \end{aligned}$$



The finite Weyl group $W_0 = \langle s_1, \ldots, s_n \rangle \simeq S_n \ltimes (\mathbb{Z}/2\mathbb{Z})^n$ is the group of signed permutations.

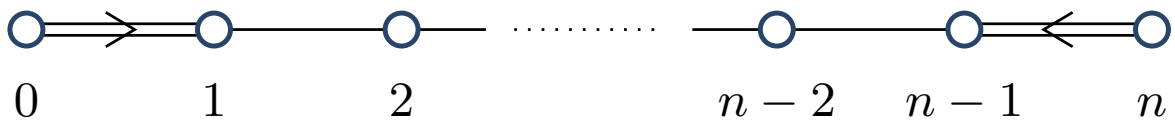
The affine Weyl group of type $\widetilde{C}_n$



W acts on Laurent polynomials in $z_1, \ldots, z_n$ by

$$
\begin{aligned}
s_i(z_1, \ldots, z_i, z_{i+1}, \ldots, z_n) &= (z_1, \ldots, z_{i+1}, z_i, \ldots, z_n), \qquad 1 \le i < n, \\
s_n(z_1, \ldots, z_{n-1}, z_n) &= (z_1, \ldots, z_{n-1}, z_n^{-1}), \\
s_0(z_1, z_2, \ldots, z_n) &= (q z_1^{-1}, z_2, \ldots, z_n).
\end{aligned}
$$

The affine Weyl group of type $\widetilde{C}_n$



W acts on Laurent polynomials in $z_1, \ldots, z_n$ by

$$\begin{aligned} s_i(z_1, \ldots, z_i, z_{i+1}, \ldots, z_n) &= (z_1, \ldots, z_{i+1}, z_i, \ldots, z_n), \qquad 1 \le i < n, \\ s_n(z_1, \ldots, z_{n-1}, z_n) &= (z_1, \ldots, z_{n-1}, z_n^{-1}), \\ s_0(z_1, z_2, \ldots, z_n) &= (qz_1^{-1}, z_2, \ldots, z_n). \end{aligned}$$

A Laurent polynomial is symmetric if it is invariant under the action of W_0.

Example: The symmetric Koornwinder polynomial $K_{(2,0)}$

$$\begin{aligned} K_{(2,0)} = z_1^2 + z_1^{-2} + z_2^2 + z_2^{-2} + \frac{(1+q)(1-t)}{1-qt}(z_1 + z_1^{-1})(z_2 + z_2^{-1}) \\ - \frac{(1+q)(1-qt^2)}{(1-qt)(1-q^2t^2abcd)}\big(a+b+c+d-qt(abc+abd+acd+bcd)\big) \\ \times (z_1 + z_1^{-1} + z_2 + z_2^{-1}) + \text{constant}. \end{aligned}$$

The affine Hecke algebra $\mathcal{H}$ of type $\widetilde{C}_n$

The affine Hecke algebra $\mathcal{H}$ is generated by $T_0, T_1, \ldots, T_n$, subject to the braid relations

$$\begin{aligned} T_i T_j &= T_j T_i, && |i-j| \geq 2, \\ T_i T_{i+1} T_i &= T_{i+1} T_i T_{i+1}, && 1 \leq i < n-1, \\ T_0 T_1 T_0 T_1 &= T_1 T_0 T_1 T_0, \\ T_{n-1} T_n T_{n-1} T_n &= T_n T_{n-1} T_n T_{n-1}, \end{aligned}$$

and the quadratic relations

$$(T_i - t_i)(T_i + 1) = 0, \qquad 0 \leq i \leq n,$$

where $t_0 = -ac/q$, $t_i = t$ for $1 \leq i < n$, and $t_n = -bd$.



When $t_i = 1$ for all $0 \leq i \leq n$, we get back the affine Weyl group W, where T_i corresponds to s_i.

Polynomial representation of $\mathcal{H}$

The affine Hecke algebra $\mathcal{H}$ acts on Laurent polynomials in $z_1, \ldots, z_n$ by the Demazure–Lusztig operators

$$T_i = t + \frac{tz_i - z_{i+1}}{z_i - z_{i+1}}\,(s_i - 1), \qquad 1 \le i < n,$$

$$T_n = -bd + \frac{(1 - bz_n)(1 - dz_n)}{1 - z_n^2}\,(s_n - 1),$$

$$T_0 = -\frac{ac}{q} + \frac{(1 - a/z_1)(1 - c/z_1)}{1 - q/z_1^2}\,(s_0 - 1).$$

Theorem (Sahi '99)

This representation is faithful.

Koornwinder ASEP polynomials

Fix a partition λ. The Koornwinder ASEP polynomials $\{F_\eta\}_{\eta \in W_0 \cdot \lambda}$ are the unique family of Laurent polynomials, normalized by $[z^\eta]F_\eta = 1$, satisfying the qKZ relations (Kasatani '11)

$$
\begin{aligned}
T_0 F_{\eta_1,\eta_2,\ldots,\eta_n} &= q^{\eta_1} F_{-\eta_1,\eta_2,\ldots,\eta_n}, && \eta_1 < 0, \\
T_0 F_{\eta_1,\eta_2,\ldots,\eta_n} &= t_0\, F_{\eta_1,\eta_2,\ldots,\eta_n}, && \eta_1 = 0, \\
T_i F_{\ldots,\eta_i,\eta_{i+1},\ldots} &= t\, F_{\ldots,\eta_i,\eta_{i+1},\ldots}, && \eta_i = \eta_{i+1},\ 1 \le i < n, \\
T_i F_{\ldots,\eta_i,\eta_{i+1},\ldots} &= F_{\ldots,\eta_{i+1},\eta_i,\ldots}, && \eta_i > \eta_{i+1},\ 1 \le i < n, \\
T_n F_{\ldots,\eta_n} &= t_n\, F_{\ldots,\eta_n}, && \eta_n = 0, \\
T_n F_{\ldots,\eta_n} &= F_{\ldots,-\eta_n}, && \eta_n > 0.
\end{aligned}
$$

Recall that $t_0 = -ac/q$ and $t_n = -bd$.